

International Islamic University Chittagong (IIUC)

Department of Computer Science and Engineering (CSE)

Course Title: Mathematics-III (Matrices, Linear System of Equation and Vector Analysis)

Course Code: MATH-2307

Decomposition Theorem

Eigen Value Decomposition Theorem

1. Define Eigen value decomposition theorem. Find the Eigen value decomposition for the matrix $A = \begin{bmatrix} 4 & 1 \\ 1 & 4 \end{bmatrix}$ with justification.
2. Find the Eigen value decomposition for the matrix $A = \begin{bmatrix} 5 & 4 \\ 1 & 2 \end{bmatrix}$ with justification.
3. Examine the Eigen decomposition for the matrix $\begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}$

LU Decomposition OR LU Factorization

1. Define LU decomposition. Solve the linear system of equations using LU decomposition,
$$\begin{aligned} x + 2y + 3z &= 5 \\ 2x - 4y + 6z &= 18 \\ 3x - 9y - 3z &= 6 \end{aligned}$$
2. Solve the linear system of equations using LU decomposition,
$$\begin{aligned} x + y + z &= 6 \\ x - y + z &= 2 \\ 2x + y - z &= 1 \end{aligned}$$

Singular Value Decomposition (SVD)

1. Define singular value decomposition. Compute the singular value decomposition for the matrix $A = \begin{bmatrix} 2 & -2 \\ 1 & 1 \end{bmatrix}$ with justification.

Vector

1. Define Vector, Scalar, Unit vector, Vector field, Scalar field.
2. If $\vec{r}_1 = 3\hat{i} - 2\hat{j} + \hat{k}$, $\vec{r}_2 = 2\hat{i} - 4\hat{j} - 3\hat{k}$, $\vec{r}_3 = -\hat{i} + 2\hat{j} + 2\hat{k}$. Find the magnitude of
 - i. $\vec{r}_3$
 - ii. $\vec{r}_1 + \vec{r}_2 + \vec{r}_3$
 - iii. $2\vec{r}_1 - 3\vec{r}_2 - 5\vec{r}_3$
3. If $\vec{r}_1 = 2\hat{i} - \hat{j} + \hat{k}$, $\vec{r}_2 = \hat{i} + 3\hat{j} - 2\hat{k}$, $\vec{r}_3 = -2\hat{i} + \hat{j} - 3\hat{k}$, $\vec{r}_4 = 3\hat{i} + 2\hat{j} + 5\hat{k}$. Find scalars a, b, c such that, $\vec{r}_4 = a\vec{r}_1 + b\vec{r}_2 + c\vec{r}_3$.
4. Find a unit vector parallel to the resultant of vectors, $\vec{r}_1 = 2\hat{i} + 4\hat{j} - 5\hat{k}$, $\vec{r}_2 = \hat{i} + 2\hat{j} + 3\hat{k}$. with justification.

The Dot and Cross Product

1. Define dot and cross product of two vectors.
2. Find the angle between the vectors, $\vec{A} = 2\hat{i} + 2\hat{j} - \hat{k}$, $\vec{B} = 6\hat{i} - 3\hat{j} + 2\hat{k}$.
3. Find the angles, which the vector $\vec{A} = 2\hat{i} - \hat{j} + 2\hat{k}$ makes with co-ordinate axes.
4. Find the projection and component of the vector, $\vec{A} = \hat{i} - 2\hat{j} + \hat{k}$ on the vector $\vec{B} = 4\hat{i} - 4\hat{j} + 7\hat{k}$.
5. Find a unit vector perpendicular to the vector $\vec{A} = 3\hat{i} + \hat{j}$ and $\vec{B} = -\hat{i} + 2\hat{j} + 2\hat{k}$.
6. Determine a unit vector perpendicular to the plane of, $\vec{A} = 2\hat{i} - 6\hat{j} - 3\hat{k}$, $\vec{B} = 4\hat{i} + 3\hat{j} - \hat{k}$.
7. Find the work done in among an object along a vector $\vec{r} = 3\hat{i} + 2\hat{j} - 5\hat{k}$ if the applied force is $\vec{F} = 2\hat{i} - \hat{j} - \hat{k}$.
8. Find the distance from the origin to the plane if the plane is perpendicular to the vector $\vec{A} = 2\hat{i} + 3\hat{j} + 6\hat{k}$ and passing through the terminal point of the vector $\vec{A} = \hat{i} + 5\hat{j} + 3\hat{k}$
9. If $\vec{A} = A_x\hat{i} + A_y\hat{j} + A_z\hat{k}$ and $\vec{B} = B_x\hat{i} + B_y\hat{j} + B_z\hat{k}$ find $\vec{A} \times \vec{B}$
10. Show that $\vec{A} = \hat{i} + 2\hat{j} - 3\hat{k}$, $\vec{B} = 2\hat{i} - \hat{j} + 2\hat{k}$ and $\vec{C} = 3\hat{i} + \hat{j} - \hat{k}$ are coplanar.
11. On which condition of λ the vectors $\hat{i} + 2\hat{j} + 3\hat{k}$, $\lambda\hat{i} + 4\hat{j} + 7\hat{k}$ and $-3\hat{i} - 2\hat{j} - 5\hat{k}$ are collinear and hence determine the vectors are coplanar or not.
12. If $\vec{a} \cdot \vec{b} = \sqrt{3}$ and $\vec{a} \times \vec{b} = \hat{i} + 2\hat{j} + 2\hat{k}$, find the angle between $\vec{a}$ and $\vec{b}$
13. If $\vec{A} = 2\hat{i} - 3\hat{j} - \hat{k}$ and $\vec{B} = \hat{i} + 4\hat{j} - 2\hat{k}$, $\vec{C} = 2\hat{i} - \hat{j} - \hat{k}$. Find
 - i. $\vec{A} \times \vec{B}$
 - ii. $\vec{B} \times \vec{A}$
 - iii. $(\vec{A} + \vec{B}) \times (\vec{A} - \vec{B})$
 - iv. $(\vec{A} \times \vec{B}) \times \vec{C}$
14. Find the area of the triangle having vertices at P(1, 3, 2), Q(2, -1, 1) and R (-1, 2, 3)
15. Evaluate $(2\hat{i} - 3\hat{j}) \cdot [(\hat{i} + \hat{j} - \hat{k}) \times (3\hat{i} - \hat{k})]$

Vector Differentiation

1. A particle moves along a curve whose parametric equations are $x = e^{-t}$, $y = 2\cos 3t$, $z = 2\sin 3t$, where t is the time
 - a. Determine its velocity and acceleration at any time.
 - b. Find the magnitudes of the velocity and acceleration at $t = 0$.
2. A particle moves through 3-space in such a way that its velocity is $\vec{V}(t) = \hat{i} + t\hat{j} + t^2\hat{k}$. Find the co-ordinates of the particle at time $t = 1$ given that the particle is at the point $(-1, 2, 4)$ at time $t = 0$
3. A particle move along the curve, $x = 2t^2$, $y = t^2 - 4t$, $z = 3t - 5$, where t is time. Find the components of its velocity and acceleration at the time $t = 1$ in the direction $\hat{i} - 3\hat{j} + 2\hat{k}$
4. Find the unit tangent (Slope/ Differentiation) vector to any point on the curve $x = t^2 + 1$, $y = 4t - 3$ and $z = 2t^2 - 6t$. Determine the unit tangent at the point where $t = 2$.
5. If $\vec{A} = 5t^2\hat{i} + t\hat{j} - t^3\hat{k}$ and $\vec{B} = \sin t\hat{i} - \cos t\hat{j}$, find
 - i. $\frac{d}{dt}(\vec{A} \cdot \vec{B})$
 - ii. $\frac{d}{dt}(\vec{A} \times \vec{B})$
 - iii. $\frac{d}{dt}(\vec{A} \cdot \vec{A})$

6. A particle moves so that its position vector is given by $\vec{r} = \cos\omega t\hat{i} + \sin\omega t\hat{j}$, where ω is a constant. Show that
- The velocity $\vec{V}$ of the particle is perpendicular to $\vec{r}$.
 - The acceleration $\vec{a}$ is directed toward the origin and has magnitude proportional to the distance from the origin.
 - $\vec{r} \times \vec{V} = a$, constant vector.
7. If $\vec{A} = (2x^2y - x^4)\hat{i} + (e^{xy} - y\sin x)\hat{j} + (x^2\cos y)\hat{k}$, find $\frac{\partial \vec{A}}{\partial x}, \frac{\partial \vec{A}}{\partial y}, \frac{\partial^2 \vec{A}}{\partial x^2}, \frac{\partial^2 \vec{A}}{\partial x \partial y}, \frac{\partial^2 \vec{A}}{\partial y \partial x}$
8. If $\phi(x, y, z) = xyz$ and $\vec{A} = xz\hat{i} - xy^2\hat{j} + yz^2\hat{k}$, find $\frac{\partial^3}{\partial x^2 \partial z}(\phi \vec{A})$ at the point $(2, -1, 1)$
9. If $\vec{A} = x^2yz\hat{i} - 2xz^3\hat{j} + xz^2\hat{k}$ and $\vec{B} = 2z\hat{i} + y\hat{j} - x^2\hat{k}$, find $\frac{\partial^2}{\partial x \partial y}((\vec{A} \times \vec{B}))$ at $(1, 0, -2)$

Gradient, Divergence and Curl

- Define gradient, divergence and curl of a vector.

Gradient

- If $\phi(x, y, z) = 3x^2y - y^3z^2$ find $\vec{\nabla}\phi$ (grad ϕ) at the point $(1, -2, -1)$
- Find $\vec{\nabla}\phi$ if (a) $\phi = \ln|\vec{r}|$ (b) $\phi = \frac{1}{r}$
- Show that $\vec{\nabla}\phi$ is a vector perpendicular to the surface, $\phi(x, y, z) = c$, where c is a constant.
- Find a unit normal to the surface $x^2y + 2xz = 4$ at the point $(2, -2, 3)$
- Find the directional derivative of $\phi = x^2yz + 4xz^2$ at $(1, -2, -1)$ in the direction $2\hat{i} - \hat{j} - 2\hat{k}$
- Find the **directional derivative** of the function $\phi = x^2y + y^2z + z^2x$ at the point $(1, -1, 2)$ in the direction of the vector $\vec{A} = 4\hat{i} + 2\hat{j} - 5\hat{k}$
- Find the angle between the surface $x^2 + y^2 + z^2 = 9$ and $z = x^2 + y^2 - 1$ at the point $(2, -1, 2)$

Divergence

- If $\vec{A} = x^2z\hat{i} - 2y^3z^2\hat{j} + xy^2z\hat{k}$, find $\vec{\nabla} \cdot \vec{A}$ ($\text{div } \vec{A}$) at the point $(1, -1, 1)$
- If $\phi(x, y, z) = 2x^3y^2z^4$, (a) Find $\vec{\nabla} \cdot \vec{\nabla}\phi$ (or $\text{div grad } \phi$) (b) Show that $\vec{\nabla} \cdot \vec{\nabla}\phi = \vec{\nabla}^2\phi$, where $\vec{\nabla}^2 = \frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + \frac{\partial^2}{\partial z^2}$
- Prove that, $\vec{\nabla}^2\left(\frac{1}{r}\right) = 0$.
- Prove that, $\vec{\nabla} \cdot \left(\frac{\vec{r}}{r^3}\right) = 0$.
- Show that, $\vec{\nabla} = \frac{\vec{r}}{r} \frac{\partial}{\partial r}$
- Determine the constant a so that the vector $\vec{V} = (x + 3y)\hat{i} + (y - 2z)\hat{j} + (x + az)\hat{k}$ is solenoidal.

Curl

1. If $\vec{A} = xz^3\hat{i} - 2x^2yz\hat{j} + 2yz^4\hat{k}$, find $\vec{\nabla} \times \vec{A}$ (or $\text{curl } \vec{A}$) at the point $(1, -1, 1)$
2. If $\vec{v} = \vec{\omega} \times \vec{r}$, prove $\vec{\omega} = \frac{1}{2} \text{curl } \vec{v}$, where $\vec{\omega}$ is a constant vector.
3. A vector $\vec{v}$ is called irrotational if $\text{curl } \vec{v} = 0$. Find constants a, b, c so that $\vec{v} = (x + 2y + az)\hat{i} + (bx - 3y - z)\hat{j} + (4x + cy + 2z)\hat{k}$ is irrotational.

Line Integral

1. If $\vec{A} = (3x^2 + 6y)\hat{i} - 14yz\hat{j} + 20xz^2\hat{k}$, evaluate $\int \vec{A} \cdot d\vec{r}$ from $(0,0,0)$ to $(1,1,1)$ along the following paths c .
 - a. $x = t, y = t^2, z = t^3$
 - b. the straight lines from $(0,0,0)$ to $(1,0,0)$ then to $(1,1,0)$ and then to $(1,1,1)$
 - c. the straight line joining $(0,0,0)$ and $(1,1,1)$
2. Find the total work done in moving a particle in a force field given by $\vec{F} = 3xy\hat{i} - 5z\hat{j} + 10x\hat{k}$ along the curve, $x = t^2 + 1, y = 2t^2, z = t^3$ from $t = 1$ to $t = 2$
3. Find the work done by the force $\vec{F} = x^3y\hat{i} - (x - y)\hat{j}$ on a particle that moves along the parabola $y = x^2$ from $(-2, 4)$ to $(1, 1)$
4. Evaluate the line integral $\int \vec{F} \cdot d\vec{r}$ where the force field is given by $\vec{F}(x, y) = 3xy\hat{i} - 5z\hat{j} + 10x\hat{k}$ along the curve $x = t^2 + 1, y = 2t^2, z = t^3$ from $t = 1$ to $t = 2$
5. Evaluate $\int xy dx$ from $B(1,0)$ to $C(0,1)$ along the curve C that is the portion of $x^2 + y^2 = 1$ in the first quadrant.
6. If $\vec{F} = 3xy\hat{i} - y^2\hat{j}$, evaluate $\int \vec{F} \cdot d\vec{r}$ where c is the curve in the xy plane, $y = 2x^2$, from $(0,0)$ to $(1,2)$
7. Find the work done in moving a particle once around a circle C in the xy plane, if the circle has center at the origin and radius 3 and if the force field is given by $\vec{F} = (2x - y + z)\hat{i} + (x + y - z^2)\hat{j} + (3x - 2y + 4z)\hat{k}$
8. If $\phi = 2xyz^2$, $\vec{F} = xy\hat{i} - z\hat{j} + x^2\hat{k}$ and c is the curve $x = t^2, y = 2t, z = t^3$ from $t = 0$ to $t = 1$ evaluate the line integral
 - i. $\int \phi \cdot d\vec{r}$
 - ii. $\int \vec{F} \times d\vec{r}$

Surface Integral and Volume Integral

1. If $\vec{F} = 2z\hat{i} - x\hat{j} + y\hat{k}$, evaluate $\int \vec{F} \cdot d\vec{v}$, where v is bounded by the surfaces $x = 0, x = 2, y = 0, y = 4, z = x^2, z = 2$
2. Evaluate $\int \vec{F} \cdot d\vec{v}$, where v is the region bounded by the planes, $x = 0, x = 2, y = 0, y = 3, z = 0, z = 4$ and $\vec{F} = xy\hat{i} + z\hat{j} - x^2\hat{k}$

3. **State Divergence theorem.** Verify the divergence theorem for the vector field $\vec{F} = x^2\hat{i} + z\hat{j} + y\hat{k}$ taken over the region bounded by the planes, $z = 0, z = 2, x = 0, x = 1, y = 0, y = 3$
4. Verify the Divergence theorem for $\vec{F} = (2xy + z)\hat{i} + y^2\hat{j} - (x + 3y)\hat{k}$ taken over the region bounded by the planes, $2x + 2y + z = 6, x = 0, y = 0, z = 0$
5. Verify the divergence theorem for the vector field $\vec{F} = 4xz\hat{i} - y^2\hat{j} + yz\hat{k}$ taken over the region bounded by the planes, $x = 0, x = 2, y = 0, y = 2, z = 0, z = 2$
6. If $\vec{F} = 4xz\hat{i} - y^2\hat{j} + yz\hat{k}$, show that, $\iint_S \vec{F} \cdot \hat{n} ds = \frac{3}{2}$ where S is the surface of the cube bounded by $x = 0, x = 1, y = 0, y = 1, z = 0, z = 1$
7. **State Green's theorem.** Verify the Green's theorem in the plane for $\oint_C \{(2xy - x^2)dx + (x + y^2)dy\}$ where C is the closed curve of the region bounded by $y = x^2$, and $y^2 = x$.
8. Verify Green's theorem in the plane for $\oint_C \{(xy + y^2)dx + x^2dy\}$ where C is the closed curve of the region bounded by $y = x$ and $y = x^2$
9. Evaluate $\int xydx + (2x - y)dy$ round the region bounded on the curve $y = x^2$ and $x = y^2$ by using Green's theorem.